



Exercise-1

Marked questions are recommended for Revision.

PART - I : SUBJECTIVE QUESTIONS

Section (A): Matrix, Algebra of Matrix, Transpose, symmetric and skew symmetric matrix,

A-1. Construct a 3×2 matrix whose elements are given by $a_{ij} = 2i - j$.

A-2. If $\begin{bmatrix} x-y & 1 & z \\ 2x-y & 0 & w \end{bmatrix} = \begin{bmatrix} -1 & 1 & 4 \\ 0 & 0 & 5 \end{bmatrix}$, find x, y, z, w .

A-3. Let $A + B + C = \begin{bmatrix} 4 & -1 \\ 0 & 1 \end{bmatrix}$, $4A + 2B + C = \begin{bmatrix} 0 & -1 \\ -3 & 2 \end{bmatrix}$ and $9A + 3B + C = \begin{bmatrix} 0 & -2 \\ 2 & 1 \end{bmatrix}$ then find A

A-4. If $A = \begin{bmatrix} 1 & 2 \\ 3 & -4 \\ 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 5 & 6 \\ 7 & -8 & 2 \end{bmatrix}$, will AB be equal to BA . Also find AB & BA .

A-5. If $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$, then show that $A^3 = \begin{bmatrix} 7 & -12 \\ 3 & -5 \end{bmatrix}$

A-6. If $A = \begin{bmatrix} 0 & -\tan \frac{\alpha}{2} \\ \tan \frac{\alpha}{2} & 0 \end{bmatrix}$ show that $(I + A) = (I - A) \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$

A-7. Given $F(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$. If $x \in \mathbb{R}$ Then for what values of y , $F(x + y) = F(x) F(y)$.

$$F(x + y) = F(x) F(y)$$

A-8. Let $A = [a_{ij}]_{n \times n}$ where $a_{ij} = i^2 - j^2$. Show that A is skew-symmetric matrix.

A-9. If $C = \begin{bmatrix} 1 & 4 & 6 \\ 7 & 2 & 5 \\ 9 & 8 & 3 \end{bmatrix} \begin{bmatrix} 0 & 2 & 3 \\ -2 & 0 & 4 \\ -3 & -4 & 0 \end{bmatrix} \begin{bmatrix} 1 & 7 & 9 \\ 4 & 2 & 8 \\ 6 & 5 & 3 \end{bmatrix}$, then trace of $C + C^3 + C^5 + \dots + C^{99}$ is

Section (B) : Determinant of Matrix

B-1. If the minor of three-one element (i.e. M_{31}) in the determinant $\begin{vmatrix} 0 & 1 & \sec \alpha \\ \tan \alpha & -\sec \alpha & \tan \alpha \\ 1 & 0 & 1 \end{vmatrix}$ is 1 then find the value of α . ($0 \leq \alpha \leq \pi$).



B-2. Using the properties of determinants, evaluate:

$$(i) \begin{vmatrix} 23 & 6 & 11 \\ 36 & 5 & 26 \\ 63 & 13 & 37 \end{vmatrix}$$

$$(ii) \begin{vmatrix} 0 & c & b \\ -c & 0 & a \\ -b & -a & 0 \end{vmatrix}$$

$$(iii) \begin{vmatrix} 103 & 115 & 114 \\ 111 & 108 & 106 \\ 104 & 113 & 116 \end{vmatrix} + \begin{vmatrix} 113 & 116 & 104 \\ 108 & 106 & 111 \\ 115 & 114 & 103 \end{vmatrix}.$$

$$(iv) \begin{vmatrix} \sqrt{13} + \sqrt{3} & 2\sqrt{5} & \sqrt{5} \\ \sqrt{15} + \sqrt{26} & 5 & \sqrt{10} \\ 3 + \sqrt{65} & \sqrt{15} & 5 \end{vmatrix}$$

B-3. Prove that :

$$(i) \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c)$$

$$(ii) \begin{vmatrix} a & b+c & a^2 \\ b & c+a & b^2 \\ c & a+b & c^2 \end{vmatrix} = -(a+b+c)(a-b)(b-c)(c-a)$$

$$(iii) \begin{vmatrix} b+c & a & a \\ b & c+a & b \\ c & c & a+b \end{vmatrix} = 4abc$$

$$(iv) \text{ If } \begin{vmatrix} 1 & a^2 & a^4 \\ 1 & b^2 & b^4 \\ 1 & c^2 & c^4 \end{vmatrix} = (a+b)(b+c)(c+a) \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix}.$$

B-4. If a, b, c are positive and are the $p^{\text{th}}, q^{\text{th}}, r^{\text{th}}$ terms respectively of a G.P., show without expanding that,

$$\begin{vmatrix} \log a & p & 1 \\ \log b & q & 1 \\ \log c & r & 1 \end{vmatrix} = 0.$$

B-5. Find the non-zero roots of the equation,

$$(i) \Delta = \begin{vmatrix} a & b & ax+b \\ b & c & bx+c \\ ax+b & bx+c & c \end{vmatrix} = 0. \quad (ii) \begin{vmatrix} 15-2x & 11 & 10 \\ 11-3x & 17 & 16 \\ 7-x & 14 & 13 \end{vmatrix} = 0$$

B-6. If $S_r = \alpha^r + \beta^r + \gamma^r$ then show that $\begin{vmatrix} S_0 & S_1 & S_2 \\ S_1 & S_2 & S_3 \\ S_2 & S_3 & S_4 \end{vmatrix} = (\alpha - \beta)^2 (\beta - \gamma)^2 (\gamma - \alpha)^2$.

B-7. Show that $\begin{vmatrix} a_1 l_1 + b_1 m_1 & a_1 l_2 + b_1 m_2 & a_1 l_3 + b_1 m_3 \\ a_2 l_1 + b_2 m_1 & a_2 l_2 + b_2 m_2 & a_2 l_3 + b_2 m_3 \\ a_3 l_1 + b_3 m_1 & a_3 l_2 + b_3 m_2 & a_3 l_3 + b_3 m_3 \end{vmatrix} = 0$.

B-8. If $\begin{vmatrix} e^x & \sin x \\ \cos x & \ln(1+x) \end{vmatrix} = A + Bx + Cx^2 + \dots$, then find the value of A and B .

Section (C) : Cofactor matrix, adj matrix and inverse of matrix

C-1. If $A = \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 5 & 2 \\ 7 & 4 \end{bmatrix}$, $C = \begin{bmatrix} 2 & 5 \\ 3 & 8 \end{bmatrix}$ and $AB - CD = 0$ find D .

- C-2.** (i) Prove that $(\text{adj adj } A) = |A|^{n-2} A$
 (ii) Find the value of $|\text{adj adj adj } A|$ in terms of $|A|$





C-3. If $A^{-1} = \begin{bmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{bmatrix}$ & $B = \begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix}$, find $(AB)^{-1}$

C-4. If A is a symmetric and B skew symmetric matrix and $(A + B)$ is non-singular and $C = (A + B)^{-1} (A - B)$, then prove that

(i) $C^T (A + B) C = A + B$ (ii) $C^T (A - B) C = A - B$

C-5. If $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & a & 1 \end{bmatrix}$, $A^{-1} = \begin{bmatrix} 1/2 & -1/2 & 1/2 \\ -4 & 3 & c \\ 5/2 & -3/2 & 1/2 \end{bmatrix}$, then find values of a & c.

Section (D) : Characteristic equation and system of equations

D-1. For the matrix $A = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$ find a & b so that $A^2 + aA + bI = 0$. Hence find A^{-1} .

D-2. Find the total number of possible square matrix A of order 3 with all real entries, whose adjoint matrix B has characteristics polynomial equation as $\lambda^3 - \lambda^2 + \lambda + 1 = 0$.

D-3. If $A = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 2 & 1 \\ 1 & 0 & 2 \end{bmatrix}$, show that $A^3 = (5A - I) (A - I)$

D-4. Apply Cramer's rule to solve the following simultaneous equations.

(i) $2x + y + 6z = 46$
 $5x - 6y + 4z = 15$
 $7x + 4y - 3z = 19$

(ii) $x + 2y + 3z = 2$
 $x - y + z = 3$
 $5x - 11y + z = 17$

D-5. Solve using Cramer's rule: $\frac{4}{x+5} + \frac{3}{y+7} = -1$ & $\frac{6}{x+5} - \frac{6}{y+7} = -5$.

D-6. Find those values of c for which the equations:

$2x + 3y = 3$
 $(c+2)x + (c+4)y = c+6$
 $(c+2)^2x + (c+4)^2y = (c+6)^2$ are consistent.

Also solve above equations for these values of c.

D-7. Solve the following systems of linear equations by matrix method.

(i) $2x - y + 3z = 8$ (ii) $x + y + z = 9$
 $-x + 2y + z = 4$ $2x + 5y + 7z = 52$
 $3x + y - 4z = 0$ $2x + y - z = 0$

D-8. Investigate for what values of λ, μ the simultaneous equations

$x + y + z = 6$; $x + 2y + 3z = 10$ & $x + 2y + \lambda z = \mu$ have;

- (a) A unique solution
 (b) An infinite number of solutions.
 (c) No solution.

D-9. Determine the product $\begin{bmatrix} -4 & 4 & 4 \\ -7 & 1 & 3 \\ 5 & -3 & -1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 1 \\ 1 & -2 & -2 \\ 2 & 1 & 3 \end{bmatrix}$ and use it to solve the system of equations $x - y + z = 4$, $x - 2y - 2z = 9$, $2x + y + 3z = 1$.



D-10. Compute A^{-1} , if $A = \begin{bmatrix} 3 & -2 & 3 \\ 2 & 1 & -1 \\ 4 & -3 & 2 \end{bmatrix}$. Hence solve the matrix equations

$$\begin{bmatrix} 3 & 0 & 3 \\ 2 & 1 & 0 \\ 4 & 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8 \\ 1 \\ 4 \end{bmatrix} + \begin{bmatrix} 2y \\ z \\ 3y \end{bmatrix}.$$

D-11. Which of the following statement(s) is/are true

$$4x - 5y - 2z = 2$$

S1 : The system of equations $5x - 4y + 2z = 3$ is Inconsistent.

$$2x + 2y + 8z = 1$$

S2 : A matrix 'A' has 6 elements. The number of possible orders of A is 6.

S3 : For any 2×2 matrix A, if $A(\text{adj}A) = \begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix}$, then $|A| = 10$.

S4 : If A is skew symmetric, then $B'AB$ is also skew symmetric.

D-12. Let A is non scalar matrix of order two satisfying the relation $A^2 + \ell A + mI = O$, then prove that characteristic equation of matrix A is $x^2 + \ell x + m = 0$.

PART - II : ONLY ONE OPTION CORRECT TYPE

Section (A): Matrix, Algebra of Matrix, Transpose, symmetric and skew symmetric matrix,

A-1. $\begin{bmatrix} x^2 + x & x \\ 3 & 2 \end{bmatrix} + \begin{bmatrix} 0 & -1 \\ -x+1 & x \end{bmatrix} = \begin{bmatrix} 0 & -2 \\ 5 & 1 \end{bmatrix}$ then x is equal to -
 (A) -1 (B) 2 (C) 1 (D) No value of x

A-2. If $A = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ and $B = \begin{bmatrix} -5 & 4 & 0 \\ 0 & 2 & -1 \\ 1 & -3 & 2 \end{bmatrix}$, then
 (A) $AB = \begin{bmatrix} -5 & 8 & 0 \\ 0 & 4 & -2 \\ 3 & -9 & 6 \end{bmatrix}$ (B) $AB = [-2 \ -1 \ 4]$
 (C) $AB = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$ (D) AB does not exist

A-3. If $AB = O$ for the matrices
 $A = \begin{bmatrix} \cos^2 \theta & \cos \theta \sin \theta \\ \cos \theta \sin \theta & \sin^2 \theta \end{bmatrix}$ and $B = \begin{bmatrix} \cos^2 \phi & \cos \phi \sin \phi \\ \cos \phi \sin \phi & \sin^2 \phi \end{bmatrix}$ then $\theta - \phi$ is
 (A) an odd multiple of $\frac{\pi}{2}$ (B) an odd multiple of π
 (C) an even multiple of $\frac{\pi}{2}$ (D) 0

A-4. If $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $J = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$, then $B =$
 (A) $I \cos \theta + J \sin \theta$ (B) $I \cos \theta - J \sin \theta$ (C) $I \sin \theta + J \cos \theta$ (D) $-I \cos \theta + J \sin \theta$



- A-5.** In an upper triangular matrix $A = [a_{ij}]_{n \times n}$ the elements $a_{ij} = 0$ for
 (A) $i < j$ (B) $i = j$ (C) $i > j$ (D) $i \leq j$
- A-6.** If $A = \text{diag}(2, -1, 3)$, $B = \text{diag}(-1, 3, 2)$, then $A^2B =$
 (A) $\text{diag}(5, 4, 11)$ (B) $\text{diag}(-4, 3, 18)$ (C) $\text{diag}(3, 1, 8)$ (D) B
- A-7.** If A is a skew-symmetric matrix, then trace of A is
 (A) 1 (B) -1 (C) 0 (D) none of these
- A-8.** Let $A = \begin{bmatrix} p & q \\ q & p \end{bmatrix}$ such that $\det(A) = r$ where p, q, r all prime numbers, then trace of A is equal to
 (A) 6 (B) 5 (C) 2 (D) 3
- A-9.** $A = \begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix}$ and $(A^8 + A^6 + A^4 + A^2 + I)V = \begin{bmatrix} 31 \\ 62 \end{bmatrix}$.
 (Where I is the (2×2) identity matrix), then the product of all elements of matrix V is
 (A) 2 (B) 1 (C) 3 (D) -2
- A-10.** Let $A = \begin{bmatrix} 3x^2 \\ 1 \\ 6x \end{bmatrix}$, $B = [a \ b \ c]$ and $C = \begin{bmatrix} (x+2)^2 & 5x^2 & 2x \\ 5x^2 & 2x & (x+1)^2 \\ 2x & (x+2)^2 & 5x^2 \end{bmatrix}$
 Where a, b, c and $x \in \mathbb{R}$, Given that $\text{tr}(AB) = \text{tr}(C)$, then the value of $(a + b + c)$.
 (A) 7 (B) 2 (C) 1 (D) 4
- A-11.** Which one of the following is Correct ?
 (i) The elements on the main diagonal of a symmetric matrix are all zero
 (ii) The elements on the main diagonal of a skew-symmetric matrix are all zero
 (iii) For any square matrix A , AA' is symmetric
 (iii) For any square matrix A , $(A + A')^2 = A^2 + (A')^2 + 2AA'$
 (A) FTTF (B) FFTF (C) FTTT (D) TFFT

Section (B) : Determinant of Matrix

- B-1.** If A and B are square matrices of order 3 such that $|A| = -1$, $|B| = 3$, then $|3AB|$ is equal to
 (A) -9 (B) -81 (C) -27 (D) 81
- B-2.** The absolute value of the determinant $\begin{vmatrix} -1 & 2 & 1 \\ 3 + 2\sqrt{2} & 2 + 2\sqrt{2} & 1 \\ 3 - 2\sqrt{2} & 2 - 2\sqrt{2} & 1 \end{vmatrix}$ is:
 (A) $16\sqrt{2}$ (B) $8\sqrt{2}$ (C) 8 (D) none
- B-3.** If α, β & γ are the roots of the equation $x^3 + px + q = 0$ then the value of the determinant $\begin{vmatrix} \alpha & \beta & \gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{vmatrix} =$
 (A) p (B) q (C) $p^2 - 2q$ (D) none
- B-4.** If $a, b, c > 0$ & $x, y, z \in \mathbb{R}$ then the determinant $\begin{vmatrix} (a^x + a^{-x})^2 & (a^x - a^{-x})^2 & 1 \\ (b^y + b^{-y})^2 & (b^y - b^{-y})^2 & 1 \\ (c^z + c^{-z})^2 & (c^z - c^{-z})^2 & 1 \end{vmatrix} =$
 (A) $a^xb^yc^z$ (B) $a^{-x}b^{-y}c^{-z}$ (C) $a^{2x}b^{2y}c^{2z}$ (D) zero



B-5. If a, b & c are non-zero real numbers then $D = \begin{vmatrix} b^2c^2 & bc & b+c \\ c^2a^2 & ca & c+a \\ a^2b^2 & ab & a+b \end{vmatrix} =$

(A) abc (B) $a^2b^2c^2$ (C) $bc+ca+ab$ (D) zero

B-6. The determinant $\begin{vmatrix} b_1+c_1 & c_1+a_1 & a_1+b_1 \\ b_2+c_2 & c_2+a_2 & a_2+b_2 \\ b_3+c_3 & c_3+a_3 & a_3+b_3 \end{vmatrix} =$

(A) $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ (B) $2 \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ (C) $3 \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ (D) none of these

B-7. If $x, y, z \in \mathbb{R}$ & $\Delta = \begin{vmatrix} x & x+y & x+y+z \\ 2x & 5x+2y & 7x+5y+2z \\ 3x & 7x+3y & 9x+7y+3z \end{vmatrix} = -16$ then value of x is

(A) -2 (B) -3 (C) 2 (D) 3

B-8. The determinant $\begin{vmatrix} \cos(\theta+\phi) & -\sin(\theta+\phi) & \cos 2\phi \\ \sin \theta & \cos \theta & \sin \phi \\ -\cos \theta & \sin \theta & \cos \phi \end{vmatrix}$ is:

(A) 0 (B) independent of θ (C) independent of ϕ (D) independent of θ & ϕ both

B-9. Let A be set of all determinants of order 3 with entries 0 or 1, B be the subset of A consisting of all determinants with value 1 and C be the subset of A consisting of all determinants with value -1 . Then

STATEMENT -1 : The number of elements in set B is equal to number of elements in set C .

and

STATEMENT-2 : $(B \cap C) \subset A$

- (A) STATEMENT-1 is true, STATEMENT-2 is true and STATEMENT-2 is correct explanation for STATEMENT-1
- (B) STATEMENT-1 is true, STATEMENT-2 is true and STATEMENT-2 is not correct explanation for STATEMENT-1
- (C) STATEMENT-1 is true, STATEMENT-2 is false
- (D) STATEMENT-1 is false, STATEMENT-2 is true
- (E) Both STATEMENTS are false

Section (C) : Cofactor matrix, adj matrix and inverse of matrix.

C-1. If $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$, then $\text{adj } A =$

(A) $\begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix}$ (B) $\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & -2 \\ -2 & -1 \end{bmatrix}$ (D) $\begin{bmatrix} -1 & 2 \\ 2 & -1 \end{bmatrix}$

C-2. Identify statements S_1, S_2, S_3 in order for true(T)/false(F)

S_1 : If $A = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$ then $\text{adj } A = A'$

S_2 : If $A = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$, then $A^{-1} = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$

S_3 : If B is a non-singular matrix and A is a square matrix, then $\det(B^{-1}AB) = \det(A)$

- (A) TTF (B) FTT (C) TFT (D) TTT



- C-3.** If A, B are two $n \times n$ non-singular matrices, then
 (A) AB is non-singular (B) AB is singular
 (C) $(AB)^{-1} = A^{-1} B^{-1}$ (D) $(AB)^{-1}$ does not exist
- C-4.** Let $A = \begin{bmatrix} 1 & 2 \\ 3 & -5 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ and X be a matrix such that $A = BX$, then X is equal to
 (A) $\frac{1}{2} \begin{bmatrix} 2 & 4 \\ 3 & -5 \end{bmatrix}$ (B) $\frac{1}{2} \begin{bmatrix} -2 & 4 \\ 3 & 5 \end{bmatrix}$ (C) $\begin{bmatrix} 2 & 4 \\ 3 & -5 \end{bmatrix}$ (D) none of these
- C-5.** Let $A = \begin{bmatrix} -1 & 2 & -3 \\ -2 & 0 & 3 \\ 3 & -3 & 1 \end{bmatrix}$ be a matrix, then $(\det A) \times (\text{adj } A^{-1})$ is equal to
 (A) $O_{3 \times 3}$ (B) I_3
 (C) $\begin{bmatrix} -1 & 2 & -3 \\ -2 & 0 & 3 \\ 3 & -3 & 1 \end{bmatrix}$ (D) $\begin{bmatrix} 3 & -3 & 1 \\ 3 & 0 & -2 \\ -1 & 2 & -3 \end{bmatrix}$
- C-6.** If D is a determinant of order three and Δ is a determinant formed by the cofactors of determinant D ; then following statement is **True/False**
 (i) $\Delta = D^2$ (ii) $D = 0$ implies $\Delta = 0$
 (iii) if $D = 27$, then Δ is perfect cube (iv) if $D = 27$, then Δ is perfect square
 (A) FTTT (B) TFTT (C) TTTT (D) TTFT

Section (D) : Characteristic equation and system of equations

- D-1.** If $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$ is a root of polynomial $x^3 - 6x^2 + 7x + k = 0$, then the value of k is
 (A) 2 (B) 4 (C) -2 (D) 1
- D-2.** If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ (where $bc \neq 0$) satisfies the equations $x^2 + k = 0$, then
 (A) $a + d = 0$ & $k = |A|$ (B) $a - d = 0$ & $k = |A|$
 (C) $a + d = 0$ & $k = -|A|$ (D) $a + d \neq 0$ & $k = |A|$
- D-3.** If the system of equations $x + 2y + 3z = 4$, $x + py + 2z = 3$, $x + 4y + \mu z = 3$ has an infinite number of solutions and solution triplet is
 (A) $p = 2$, $\mu = 3$ and $(5 - 4\lambda, \lambda - 1, \lambda)$ (B) $p = 2$, $\mu = 4$ and $(5 - 4\lambda, \frac{\lambda - 1}{2}, 2\lambda)$
 (C) $3p = 2\mu$ and $(5 - 4\lambda, \lambda - 1, 2\lambda)$ (D) $p = 4$, $\mu = 2$ and $(5 - 4\lambda, \frac{\lambda - 1}{2}, \lambda)$
- D-4.** Let λ and α be real. Find the set of all values of λ for which the system of linear equations have infinite solution $\forall$ real values of α .
 $\lambda x + (\sin \alpha)y + (\cos \alpha)z = 0$
 $x + (\cos \alpha)y + (\sin \alpha)z = 0$
 $-x + (\sin \alpha)y + (\cos \alpha)z = 0$
 (A) $(-\infty, \sqrt{2}) \cup (\sqrt{2}, \infty)$ (B) -1
 (C) $(-5, -\sqrt{2})$ (D) None of these
- D-5.** The value of '2k' for which the set of equations $3x + ky - 2z = 0$, $x + ky + 3z = 0$, $2x + 3y - 4z = 0$ has a non-trivial solution over the set of rational is:
 (A) 31 (B) 32 (C) 33 (D) 34





PART - III : MATCH THE COLUMN

1. Column I

(A) $\begin{bmatrix} 1 & x & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 3 & 2 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = 0$, then $x =$

(B) If A is a square matrix of order 3×3 and k is a scalar, then $\text{adj}(kA) = k^m \text{adj } A$, then m is

(C) If $A = \begin{bmatrix} 2 & \mu \\ \mu^2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} \gamma & 7 \\ 49 & \delta \end{bmatrix}$ here $(A - B)$ is upper triangular matrix then number of possible values of μ are

(D) If $\begin{vmatrix} (b+c)^2 & a^2 & a^2 \\ b^2 & (c+a)^2 & b^2 \\ c^2 & c^2 & (a+b)^2 \end{vmatrix} = k abc (a+b+c)^3$

then the value of k is

Column II

(p) 2

(q) -2

(r) 1

(s) $-\frac{9}{8}$

2. Column - I

(A) If A and B are square matrices of order 3×3 , where $|A| = 2$ and $|B| = 1$, then $|(A^{-1}) \cdot \text{adj}(B^{-1}) \cdot \text{adj}(2A^{-1})| =$

(B) If A is a square matrix such that $A^2 = A$ and $(I + A)^3 = I + kA$, then k is equal to

(C) Matrix $\begin{bmatrix} a & b & (a\alpha - b) \\ b & c & (b\alpha - c) \\ 2 & 1 & 0 \end{bmatrix}$ is non invertible ($b^2 \neq ac$) if -2α is

(D) If $A = [a_{ij}]_{3 \times 3}$ is a scalar matrix with $a_{11} = a_{22} = a_{33} = 2$ and $A(\text{adj}A) = kI_3$ then k is

Column - II

(p) 7

(q) 8

(r) 0

(s) -1

Exercise-2

Marked questions are recommended for Revision.

PART - I : ONLY ONE OPTION CORRECT TYPE

1. Two matrices A and B have in total 6 different elements (none repeated). How many different matrices A and B are possible such that product AB is defined.

- (A) $5(6!)$ (B) $3(6!)$ (C) $12(6!)$ (D) $8(6!)$

2. If $X = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$, then value of X^n is, (where n is natural number)

- (A) $\begin{bmatrix} 3n & -4n \\ n & -n \end{bmatrix}$ (B) $\begin{bmatrix} 2+n & 5-n \\ n & -n \end{bmatrix}$ (C) $\begin{bmatrix} 3^n & (-4)^n \\ 1^n & (-1)^n \end{bmatrix}$ (D) $\begin{bmatrix} 2n+1 & -4n \\ n & -(2n-1) \end{bmatrix}$

3. If A and B are two matrices such that $AB = B$ and $BA = A$, then $A^2 + B^2 =$

- (A) $2AB$ (B) $2BA$ (C) $A + B$ (D) AB

4. Find number of all possible ordered sets of two $(n \times n)$ matrices A and B for which $AB - BA = I$

- (A) infinite (B) n^2 (C) $n!$ (D) zero



5. If B, C are square matrices of order n and if $A = B + C$, $BC = CB$, $C^2 = O$, then which of following is true for any positive integer N.
 (A) $A^{N+1} = B^N (B + (N+1)C)$ (B) $A^N = B^N (B + (N+1)C)$
 (C) $A^{N+1} = B (B + (N+1)C)$ (D) $A^{N+1} = B^N (B + (N+2)C)$
6. How many 3×3 skew symmetric matrices can be formed using numbers $-2, -1, 1, 2, 3, 4, 0$ (any number can be used any number of times but 0 can be used at most 3 times)
 (A) 8 (B) 27 (C) 64 (D) 54
7. If A is a skew - symmetric matrix and n is an even positive integer, then A^n is
 (A) a symmetric matrix (B) a skew-symmetric matrix
 (C) a diagonal matrix (D) none of these
8. Number of 3×3 non symmetric matrix A such that $A^T = A^2 - I$ and $|A| \neq 0$, equals to
 (A) 0 (B) 2 (C) 4 (D) Infinite
9. Matrix A is such that $A^2 = 2A - I$, where I is the identity matrix. Then for $n \geq 2$, $A^n =$
 (A) $nA - (n-1)I$ (B) $nA - I$ (C) $2^{n-1}A - (n-1)I$ (D) $2^{n-1}A - I$

10. If $P = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$, $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $Q = PAP^T$ and $x = P^T Q^{2005} P$, then x is equal to

- (A) $\begin{bmatrix} 1 & 2005 \\ 0 & 1 \end{bmatrix}$ (B) $\begin{bmatrix} 4 + 2005\sqrt{3} & 6015 \\ 2005 & 4 - 2005\sqrt{3} \end{bmatrix}$
 (C) $\frac{1}{4} \begin{bmatrix} 2 + \sqrt{3} & 1 \\ -1 & 2 - \sqrt{3} \end{bmatrix}$ (D) $\frac{1}{4} \begin{bmatrix} 2005 & 2 - \sqrt{3} \\ 2 + \sqrt{3} & 2005 \end{bmatrix}$

11. Let $\Delta = \begin{vmatrix} \sin \theta \cos \phi & \sin \theta \sin \phi & \cos \theta \\ \cos \theta \cos \phi & \cos \theta \sin \phi & -\sin \theta \\ -\sin \theta \sin \phi & \sin \theta \cos \phi & 0 \end{vmatrix}$, then

- (A) Δ is independent of θ (B) Δ is independent of ϕ
 (C) Δ is a constant (D) none of these

12. $\Delta = \begin{vmatrix} 1+a^2+a^4 & 1+ab+a^2b^2 & 1+ac+a^2c^2 \\ 1+ab+a^2b^2 & 1+b^2+b^4 & 1+bc+b^2c^2 \\ 1+ac+a^2c^2 & 1+bc+b^2c^2 & 1+c^2+c^4 \end{vmatrix}$ is equal to

- (A) $(a-b)^2(b-c)^2(c-a)^2$ (B) $2(a-b)(b-c)(c-a)$
 (C) $4(a-b)(b-c)(c-a)$ (D) $(a+b+c)^3$

13. If $D = \begin{vmatrix} a^2+1 & ab & ac \\ ba & b^2+1 & bc \\ ca & cb & c^2+1 \end{vmatrix}$ then D =

- (A) $1 + a^2 + b^2 + c^2$ (B) $a^2 + b^2 + c^2$ (C) $(a+b+c)^2$ (D) none

14. Value of the $\Delta = \begin{vmatrix} a^3-x & a^4-x & a^5-x \\ a^5-x & a^6-x & a^7-x \\ a^7-x & a^8-x & a^9-x \end{vmatrix}$ is

- (A) 0 (B) $(a^3-1)(a^6-1)(a^9-1)$
 (C) $(a^3+1)(a^6+1)(a^9+1)$ (D) $a^{15}-1$





15. If $\Delta_1 = \begin{vmatrix} 2a & b & e \\ 2d & e & f \\ 4x & 2y & 2z \end{vmatrix}$, $\Delta_2 = \begin{vmatrix} f & 2d & e \\ 2z & 4x & 2y \\ e & 2a & b \end{vmatrix}$, then the value of $\Delta_1 - \Delta_2$ is

- (A) $x + \frac{y}{2} + z$ (B) 2 (C) 0 (D) 3

16. From the matrix equation $AB = AC$, we conclude $B = C$ provided:

- (A) A is singular (B) A is non-singular (C) A is symmetric (D) A is a square

17. Let $A = \begin{bmatrix} -2 & 7 & \sqrt{3} \\ 0 & 0 & -2 \\ 0 & 2 & 0 \end{bmatrix}$ and $A^4 = \lambda \cdot I$, then λ is

- (A) -16 (B) 16 (C) 8 (D) -8

18. If A is 3×3 square matrix whose characteristic polynomial equations is $\lambda^3 - 3\lambda^2 + 4 = 0$ then trace of $\text{adj}A$ is

- (A) 0 (B) 3 (C) 4 (D) -3

19. If a, b, c are non zeros, then the system of equations

$$(\alpha + a)x + \alpha y + \alpha z = 0$$

$$\alpha x + (\alpha + b)y + \alpha z = 0$$

$$\alpha x + \alpha y + (\alpha + c)z = 0$$

has a non-trivial solution if

- (A) $\alpha^{-1} = -(a^{-1} + b^{-1} + c^{-1})$ (B) $\alpha^{-1} = a + b + c$
(C) $\alpha + a + b + c = 1$ (D) none of these

20. **STATEMENT-1** : If $A = \begin{bmatrix} a^2 + x^2 & ab - cx & ac + bx \\ ab + xc & b^2 + x^2 & bc - ax \\ ac - bx & bc + ax & c^2 + x^2 \end{bmatrix}$ and $B = \begin{bmatrix} x & c & -b \\ -c & x & a \\ b & -a & x \end{bmatrix}$, then $|A| = |B|^2$.

STATEMENT-2 : If A^c is cofactor matrix of a square matrix A of order n then $|A^c| = |A|^{n-1}$.

- (A) STATEMENT-1 is true, STATEMENT-2 is true and STATEMENT-2 is correct explanation for STATEMENT-1
(B) STATEMENT-1 is true, STATEMENT-2 is true and STATEMENT-2 is not correct explanation for STATEMENT-1
(C) STATEMENT-1 is true, STATEMENT-2 is false
(D) STATEMENT-1 is false, STATEMENT-2 is true
(E) Both STATEMENTS are false

21. Let $A = \begin{bmatrix} a & 0 & b \\ 1 & e & 1 \\ c & 0 & d \end{bmatrix}$ $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ where $a, b, c, d, e \in \{0, 1\}$

then number of such matrix A for which system of equation $AX = O$ have unique solution.

- (A) 16 (B) 6 (C) 5 (D) none

22. If the system of equations $ax + y + z = 0$, $x + by + z = 0$ and $x + y + cz = 0$, where

$a, b, c \neq 1$, has a non-trivial solution, then the value of $\frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c}$ is

- (A) 1 (B) 2 (C) 3 (D) 4



PART-II: NUMERICAL VALUE QUESTIONS

INSTRUCTION :

- ❖ The answer to each question is **NUMERICAL VALUE** with two digit integer and decimal upto two digit.
- ❖ If the numerical value has more than two decimal places **truncate/round-off** the value to **TWO** decimal placed.

1. Let X be the solution set of the equation

$$A^x = I, \text{ where } A = \begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix} \text{ and } I \text{ is the unit matrix and } X \subset \mathbb{N} \text{ then the minimum value of}$$

$$\sum_x (\cos^x \theta + \sin^x \theta), \theta \in \mathbb{R} \text{ is :}$$

2. If A is a diagonal matrix of order 3×3 is commutative with every square matrix of order 3×3 under multiplication and $\text{tr}(A) = 10$, then the value of $|A|$ is :

3. A , is a (3×3) diagonal matrix having integral entries such that $\det(A) = 120$, number of such matrices is $10n$. Then n is :

4. If $\begin{vmatrix} b+c & c+a & a+b \\ c+a & a+b & b+c \\ a+b & b+c & c+a \end{vmatrix} \geq 0$, where $a, b, c \in \mathbb{R}^+$, then $\frac{c}{a+b}$ is

5. If $a_1, a_2, a_3, 5, 4, a_6, a_7, a_8, a_9$ are in H.P. and $D = \begin{vmatrix} a_1 & a_2 & a_3 \\ 5 & 4 & a_6 \\ a_7 & a_8 & a_9 \end{vmatrix}$, then the value of D is

6. If $\begin{vmatrix} a+b+2c & a & b \\ c & b+c+2a & b \\ c & a & c+a+2b \end{vmatrix} = k(\alpha a + \beta b + \gamma c)^3$, then $(2\alpha + \beta + \gamma)^k$ is $(\alpha, \beta, \gamma, k \in \mathbb{Z}^+)$

7. If A is a square matrix of order 3 and A' denotes transpose of matrix A , $A' A = I$ and $\det A = 1$, then $\det (A - I)$ must be equal to

8. Suppose A is a matrix such that $A^2 = A$ and $(I + A)^6 = I + kA$, then k is

9. If $\begin{vmatrix} -bc & b^2 + bc & c^2 + bc \\ a^2 + ac & -ac & c^2 + ac \\ a^2 + ab & b^2 + ab & -ab \end{vmatrix} = 64$, then $(ab + bc + ac)$ is :

10. Let $f(x) = \begin{vmatrix} 1 + \sin^2 x & \cos^2 x & 4 \sin 2x \\ \sin^2 x & 1 + \cos^2 x & 4 \sin 2x \\ \sin^2 x & \cos^2 x & 1 + 4 \sin 2x \end{vmatrix}$ then value of $f(\pi/12) + f(\pi/6) + f(\pi/4) + f(\pi/3) + f(\pi/2)$ is

11. If $U_n = \begin{vmatrix} n & 1 & 5 \\ n^2 & 2N+1 & 2N+1 \\ n^3 & 3N^2 & 3N+1 \end{vmatrix}$ and $\sum_{n=1}^N U_n = \lambda \sum_{n=1}^N n^2$, then λ is

12. The absolute value of a for which system of equations, $a^3x + (a+1)^3y + (a+2)^3z = 0$, $ax + (a+1)y + (a+2)z = 0$, $x + y + z = 0$, has a non-zero solution is:





13. Consider the system of linear equations in x, y, z :

$$(\sin 3\theta)x - y + z = 0$$

$$(\cos 2\theta)x + 4y + 3z = 0$$

$$2x + 7y + 7z = 0$$

sum of all possible values of $\theta \in (0, \pi)$ for which this system has non-trivial solution, is

14. $A_1 = [a_1]$

$$A_2 = \begin{bmatrix} a_2 & a_3 \\ a_4 & a_5 \end{bmatrix}$$

$$A_3 = \begin{bmatrix} a_6 & a_7 & a_8 \\ a_9 & a_{10} & a_{11} \\ a_{12} & a_{13} & a_{14} \end{bmatrix} \dots \dots \dots A_n = [\dots \dots \dots]$$

Where $a_r = [\log_2 r]$ ($[.]$ denotes greatest integer). Then trace of A_{10}

15. If $\left\{ \frac{1}{2}(A - A' + I) \right\}^{-1} = \frac{2}{\lambda} \begin{bmatrix} \lambda - 13 & -\frac{\lambda}{3} & \frac{\lambda}{3} \\ -17 & 10 & -1 \\ 7 & -11 & 5 \end{bmatrix}$ for $A = \begin{bmatrix} -2 & 3 & 4 \\ 5 & -4 & -3 \\ 7 & 2 & 9 \end{bmatrix}$, then λ is :

16. Given $A = \begin{bmatrix} 2 & 0 & -\alpha \\ 5 & \alpha & 0 \\ 0 & \alpha & 3 \end{bmatrix}$ For $\alpha \in \mathbb{R} - \{a, b\}$, A^{-1} exists and $A^{-1} = A^2 - 5bA + cI$, when $\alpha = 1$. The value of $a + b + c$ is :

17. Let a, b, c positive numbers. Find the number of solution of system of equations in x, y and z
 $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$; $\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$; $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ has finitely many solutions

PART - III : ONE OR MORE THAN ONE OPTIONS CORRECT TYPE

1. Suppose a_1, a_2, a_3 are in A.P. and b_1, b_2, b_3 are in H.P. and let

$$\Delta = \begin{vmatrix} a_1 - b_1 & a_1 - b_2 & a_1 - b_3 \\ a_2 - b_1 & a_2 - b_2 & a_2 - b_3 \\ a_3 - b_1 & a_3 - b_2 & a_3 - b_3 \end{vmatrix}, \text{ then}$$

- (A) Δ is independent of a_1, a_2, a_3 , (B) $a_1 - \Delta, a_2 - 2\Delta, a_3 - 3\Delta$ are in A.P.
 (C) $b_1 + \Delta, b_2 + \Delta^2, b_3 + \Delta$ are in H.P. (D) Δ is independent of b_1, b_2, b_3

2. Let $\theta = \frac{\pi}{5}$, $X = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$, O is null matrix and I is an identity matrix of order 2×2 , and if $I + X + X^2 + \dots + X^n = O$, then n can be
 (A) 9 (B) 19 (C) 4 (D) 29

3. If $\Delta = \begin{vmatrix} x & 2y - z & -z \\ y & 2x - z & -z \\ y & 2y - z & 2x - 2y - z \end{vmatrix}$, then
 (A) $x - y$ is a factor of Δ (B) $(x - y)^2$ is a factor of Δ
 (C) $(x - y)^3$ is a factor of Δ (D) Δ is independent of z





4. Let $a, b > 0$ and $\Delta = \begin{vmatrix} -x & a & b \\ b & -x & a \\ a & b & -x \end{vmatrix}$, then
- (A) $a + b - x$ is a factor of Δ (B) $x^2 + (a + b)x + a^2 + b^2 - ab$ is a factor of Δ
 (C) $\Delta = 0$ has three real roots if $a = b$ (D) $a + b + x$ is a factor of Δ
5. The determinant $\Delta = \begin{vmatrix} b & c & b\alpha + c \\ c & d & c\alpha + d \\ b\alpha + c & c\alpha + d & a\alpha^3 - c\alpha \end{vmatrix}$ is equal to zero if
- (A) b, c, d are in A.P. (B) b, c, d are in G.P.
 (C) b, c, d are in H.P. (D) α is a root of $ax^3 - bx^2 - 3cx - d = 0$
6. The determinant $\Delta = \begin{vmatrix} a^2(1+x) & ab & ac \\ ab & b^2(1+x) & bc \\ ac & bc & c^2(1+x) \end{vmatrix}$ is divisible by
- (A) $x + 3$ (B) $(1+x)^2$ (C) x^2 (D) $x^2 + 1$
7. If A is a non-singular matrix and A^T denotes the transpose of A , then:
- (A) $|A| \neq |A^T|$ (B) $|A \cdot A^T| = |A|^2$ (C) $|A^T \cdot A| = |A^T|^2$ (D) $|A| + |A^T| \neq 0$
8. Let $f(x) = \begin{vmatrix} 2\sin x & \sin^2 x & 0 \\ 1 & 2\sin x & \sin^2 x \\ 0 & 1 & 2\sin x \end{vmatrix}$, then
- (A) $f(x)$ is independent of x (B) $f'(\pi/2) = 0$
 (C) $\int_{-\pi/2}^{\pi/2} f(x)dx = 0$ (D) tangent to the curve $y = f(x)$ at $x = 0$ is $y = 0$
9. Let $f(x) = \begin{vmatrix} 1/x & \log x & x^n \\ 1 & -1/n & (-1)^n \\ 1 & a & a^2 \end{vmatrix}$, then (where $f^n(x)$ denotes n^{th} derivative of $f(x)$)
- (A) $f^n(1)$ is independent of a
 (B) $f^n(1)$ is independent of n
 (C) $f^n(1)$ depends on a and n
 (D) $y = a(x - f^n(1))$ represents a straight line through the origin
10. Let A, B, C, D be real matrices such that $A^T = BCD$; $B^T = CDA$; $C^T = DAB$ and $D^T = ABC$ for the matrix $M = ABCD$, then find M^{2016} ?
- (A) M (B) M^2 (C) M^3 (D) M^4
11. Let A and B be two 2×2 matrix with real entries. If $AB = O$ and $\text{tr}(A) = \text{tr}(B) = 0$ then
- (A) A and B are commutative w.r.t. operation of multiplication.
 (B) A and B are not commutative w.r.t. operation of multiplication.
 (C) A and B are both null matrices.
 (D) $BA = O$
12. If $A^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & -2 & 1 \\ 0 & 0 & -1 \end{bmatrix}$, then
- (A) $|A| = 2$ (B) A is non-singular
 (C) $\text{Adj. } A = \begin{bmatrix} 1/2 & -1/2 & 0 \\ 0 & -1 & 1/2 \\ 0 & 0 & -1/2 \end{bmatrix}$ (D) A is skew symmetric matrix



13. If A and B are square matrices of order 3, then the true statement is/are (where I is unit matrix).
 (A) $\det(-A) = -\det A$
 (B) If AB is singular then atleast one of A or B is singular
 (C) $\det(A + I) = 1 + \det A$
 (D) $\det(2A) = 2^3 \det A$
14. Let **M** be a 3×3 non-singular matrix with $\det(\mathbf{M}) = 4$. If $\mathbf{M}^{-1} \text{adj}(\text{adj } \mathbf{M}) = \mathbf{k}^2 \mathbf{I}$, then the value of '**k**' may be:
 (A) +2 (B) 4 (C) -2 (D) -4
15. If $\mathbf{AX} = \mathbf{B}$ where A is 3×3 and X and B are 3×1 matrices then which of the following is correct?
 (A) If $|\mathbf{A}| = 0$ then $\mathbf{AX} = \mathbf{B}$ has infinite solutions
 (B) If $\mathbf{AX} = \mathbf{B}$ has infinite solutions then $|\mathbf{A}| = 0$
 (C) If $(\text{adj}(\mathbf{A})) \mathbf{B} = 0$ and $|\mathbf{A}| \neq 0$ then $\mathbf{AX} = \mathbf{B}$ has unique solution
 (D) If $(\text{adj}(\mathbf{A})) \mathbf{B} \neq 0$ & $|\mathbf{A}| = 0$ then $\mathbf{AX} = \mathbf{B}$ has no solution

16. Let $\mathbf{M} = \begin{bmatrix} 0 & 1 & b \\ 1 & 2 & d \\ a & c & 1 \end{bmatrix}$ and $\text{adj } \mathbf{M} = \begin{bmatrix} -1 & 1 & -1 \\ 8 & -6 & 2 \\ -5 & 3 & e \end{bmatrix}$

than which of the following option is/are correct ?

- (A) $a + b + c + d + e = 8$ (B) $|\mathbf{M}| = -2$
 (C) $a = 3$ (D) $b = 3$

17. Let $\mathbf{P} = \begin{bmatrix} 2 & 3 & 6 \\ 6 & 2 & -3 \\ a & b & c \end{bmatrix}$, $\mathbf{P} \mathbf{P}^T$ is diagonal matrix, Trace of $\mathbf{P}^T \mathbf{A} \mathbf{P} = 49$, where in matrix $\mathbf{A} = [a_{ij}]$,

$a_{11} + a_{22} = 0$, $a_{33} = 1$, then which of the following statement is/are correct ?

- (A) $|\text{Det}(\mathbf{P})| = 343$ (B) $|a| + |b| + |c| = 11$
 (C) $\text{Tr}(\mathbf{P} \mathbf{P}^T) = 147$ (D) $\mathbf{A} \mathbf{P} \mathbf{P}^T$ is diagonal matrix

18. If $\mathbf{A} = \begin{bmatrix} a & 1 & 0 \\ 1 & b & d \\ 1 & b & c \end{bmatrix}$, $\mathbf{B} = \begin{bmatrix} a & 1 & 1 \\ 0 & d & c \\ f & g & h \end{bmatrix}$, $\mathbf{U} = \begin{bmatrix} f \\ g \\ h \end{bmatrix}$, $\mathbf{V} = \begin{bmatrix} a^2 \\ 0 \\ 0 \end{bmatrix}$, $\mathbf{X} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

and $\mathbf{AX} = \mathbf{U}$ has infinitely many solution. Then which of the following statement is/are correct

- (A) $\mathbf{BX} = \mathbf{V}$ has unique solution (B) $\mathbf{BX} = \mathbf{V}$ has no unique solution
 (C) if $a \neq 0$, then $\mathbf{BX} = \mathbf{V}$ has no solution. (D) if $a \neq 0$, then $\mathbf{BX} = \mathbf{V}$ has unique solution.

PART - IV : COMPREHENSION

Comprehension # 1

Let $\mathcal{A}$ be the set of all 3×3 symmetric matrices whose entries are 1,1,1,0,0,0,-1, -1, -1. B is one of the matrix in set $\mathcal{A}$ and

$$\mathbf{X} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad \mathbf{U} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \mathbf{V} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

1. Number of such matrices B in set $\mathcal{A}$ is λ , then λ lies in the interval
 (A) (30, 40) (B) (38, 40) (C) (34, 38) (D) (25, 35)
2. Number of matrices B such that equation $\mathbf{BX} = \mathbf{U}$ has infinite solutions
 (A) is at least 6 (B) is not more than 10 (C) lie between 8 to 16 (D) is zero.





3. The equation $BX = V$
 (A) is inconsistent for atleast 3 matrices B.
 (B) is inconsistent for all matrices B.
 (C) is inconsistent for at most 12 matrices B.
 (D) has infinite number of solutions for at least 3 matrices B.

Comprehension # 2

Some special square matrices are defined as follows :

Nilpotent matrix : A square matrix A is said to be nilpotent (of order 2) if, $A^2 = O$. A square matrix is said to be nilpotent of order p, if p is the least positive integer such that $A^p = O$.

Idempotent matrix : A square matrix A is said to be idempotent if, $A^2 = A$.

e.g. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ is an idempotent matrix.

Involutory matrix : A square matrix A is said to be involutory if $A^2 = I$, I being the identity matrix.

e.g. $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ is an involutory matrix.

Orthogonal matrix : A square matrix A is said to be an orthogonal matrix if $A' A = I = AA'$.

4. If A and B are two square matrices such that $AB = A$ & $BA = B$, then A & B are
 (A) Idempotent matrices (B) Involutory matrices
 (C) Orthogonal matrices (D) Nilpotent matrices

5. If the matrix $\begin{bmatrix} 0 & 2\beta & \gamma \\ \alpha & \beta & -\gamma \\ \alpha & -\beta & \gamma \end{bmatrix}$ is orthogonal, then

(A) $\alpha = \pm \frac{1}{\sqrt{2}}$ (B) $\beta = \pm \frac{1}{\sqrt{6}}$ (C) $\gamma = \pm \frac{1}{\sqrt{3}}$ (D) all of these

6. The matrix $A = \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$ is

(A) idempotent matrix (B) involutory matrix
 (C) nilpotent matrix (D) none of these

Exercise-3

Marked questions are recommended for Revision.

* Marked Questions may have more than one correct option.

PART - I : JEE (ADVANCED) / IIT-JEE PROBLEMS (PREVIOUS YEARS)

Comprehension (Q. No. 1 to 3)

Let p be an odd prime number and T_p be the following set of 2×2 matrices :

$$T_p = \left\{ A = \begin{bmatrix} a & b \\ c & a \end{bmatrix} : a, b, c \in \{0, 1, 2, \dots, p-1\} \right\}$$



- The number of A in T_p such that A is either symmetric or skew-symmetric or both, and $\det(A)$ divisible by p is
 (A) $(p-1)^2$ (B) $2(p-1)$ (C) $(p-1)^2 + 1$ (D) $2p-1$ [IIT-JEE 2010, Paper-1, (3, -1), 84]
- The number of A in T_p such that the trace of A is not divisible by p but $\det(A)$ is divisible by p is
 [Note : The trace of matrix is the sum of its diagonal entries.] [IIT-JEE 2010, Paper-1, (3, -1), 84]
 (A) $(p-1)(p^2-p+1)$ (B) $p^3-(p-1)^2$ (C) $(p-1)^2$ (D) $(p-1)(p^2-2)$
- The number of A in T_p such that $\det(A)$ is not divisible by p is [IIT-JEE 2010, Paper-1, (3, -1), 84]
 (A) $2p^2$ (B) p^3-5p (C) p^3-3p (D) p^3-p^2
- The number of all possible values of θ , where $0 < \theta < \pi$, for which the system of equations

$$(y+z)\cos 3\theta = (xyz)\sin 3\theta$$

$$x\sin 3\theta = \frac{2\cos 3\theta}{y} + \frac{2\sin 3\theta}{z}$$

$$(xyz)\sin 3\theta = (y+2z)\cos 3\theta + y\sin 3\theta$$

 have a solution (x_0, y_0, z_0) with $y_0 z_0 \neq 0$, is [IIT-JEE 2010, Paper-1, (3, 0), 84]
- Let k be a positive real number and let [IIT-JEE 2010, Paper-2, (3, 0), 79]

$$A = \begin{bmatrix} 2k-1 & 2\sqrt{k} & 2\sqrt{k} \\ 2\sqrt{k} & 1 & -2k \\ -2\sqrt{k} & 2k & -1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 0 & 2k-1 & \sqrt{k} \\ 1-2k & 0 & 2\sqrt{k} \\ -\sqrt{k} & -2\sqrt{k} & 0 \end{bmatrix}$$

 If $\det(\text{adj } A) + \det(\text{adj } B) = 10^6$, then $[k]$ is equal to
 (Note : $\text{adj } M$ denotes the adjoint of a square matrix M and $[k]$ denotes the largest integer less than or equal to k).
- Let M and N be two $2n \times 2n$ non-singular skew-symmetric matrices such that $MN = NM$. If P^T denotes the transpose of P , then $M^2 N^2 (M^T N)^{-1} (MN^{-1})^T$ is equal to [IIT-JEE 2011, Paper-1, (4, 0), 80]
 (A) M^2 (B) $-N^2$ (C) $-M^2$ (D) MN

Comprehension (7 to 9)

Let a, b and c be three real numbers satisfying

$$[a \ b \ c] \begin{bmatrix} 1 & 9 & 7 \\ 8 & 2 & 7 \\ 7 & 3 & 7 \end{bmatrix} = [0 \ 0 \ 0] \quad \dots\dots\dots(E)$$

- If the point $P(a, b, c)$, with reference to (E), lies on the plane $2x + y + z = 1$, then the value of $7a + b + c$ is [IIT-JEE 2011, Paper-1, (3, -1), 80]
 (A) 0 (B) 12 (C) 7 (D) 6
- Let ω be a solution of $x^3 - 1 = 0$ with $\text{Im}(\omega) > 0$. if $a = 2$ with b and c satisfying (E), then the value of $\frac{3}{\omega^a} + \frac{1}{\omega^b} + \frac{3}{\omega^c}$ is equal to [IIT-JEE 2011, Paper-1, (3, -1), 80]
 (A) -2 (B) 2 (C) 3 (D) -3
- Let $b = 6$, with a and c satisfying (E). If α and β are the roots of the quadratic equation $ax^2 + bx + c = 0$, then $\sum_{n=0}^{\infty} \left(\frac{1}{\alpha} + \frac{1}{\beta} \right)^n$ is [IIT-JEE 2011, Paper-1, (3, -1), 80]
 (A) 6 (B) 7 (C) $\frac{6}{7}$ (D) ∞



10. Let $\omega \neq 1$ be a cube root of unity and S be the set of all non-singular matrices of the form, $\begin{bmatrix} 1 & a & b \\ \omega & 1 & c \\ \omega^2 & \omega & 1 \end{bmatrix}$ where each of a , b and c is either ω or ω^2 . Then the number of distinct matrices in the set S is
[IIT-JEE 2011, Paper-2, (3, -1), 80]
 (A) 2 (B) 6 (C) 4 (D) 8
11. Let M be a 3×3 matrix satisfying $M \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix}$, $M \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$, and $M \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 12 \end{bmatrix}$. Then the sum of the diagonal entries of M is
[IIT-JEE 2011, Paper-2, (4, 0), 80]
12. Let $P = [a_{ij}]$ be a 3×3 matrix and let $Q = [b_{ij}]$, where $b_{ij} = 2^{i+j}a_{ij}$ for $1 \leq i, j \leq 3$. If the determinant of P is 2, then the determinant of the matrix Q is
[IIT-JEE 2012, Paper-1, (3, -1), 70]
 (A) 2^{10} (B) 2^{11} (C) 2^{12} (D) 2^{13}
13. If P is a 3×3 matrix such that $P^T = 2P + I$, where P^T is the transpose of P and I is the 3×3 identity matrix, then there exists a column matrix $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ such that
[IIT-JEE 2012, Paper-2, (3, -1), 66]
 (A) $PX = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ (B) $PX = X$ (C) $PX = 2X$ (D) $PX = -X$
- 14*. If the adjoint of a 3×3 matrix P is $\begin{bmatrix} 1 & 4 & 4 \\ 2 & 1 & 7 \\ 1 & 1 & 3 \end{bmatrix}$, then the possible value(s) of the determinant of P is (are)
[IIT-JEE 2012, Paper-2, (4, 0), 66]
 (A) -2 (B) -1 (C) 1 (D) 2
- 15.* For 3×3 matrices M and N , which of the following statement(s) is (are) **NOT** correct ?
[JEE (Advanced) 2013, Paper-1, (4, -1)/60]
 (A) $N^T M N$ is symmetric or skew symmetric, according as M is symmetric or skew symmetric
 (B) $M N - N M$ is skew symmetric for all symmetric matrices M and N
 (C) $M N$ is symmetric for all symmetric matrices M and N
 (D) $(\text{adj } M)(\text{adj } N) = \text{adj}(MN)$ for all invertible matrices M and N
- 16*. Let M be a 2×2 symmetric matrix with integer entries. Then M is invertible if
[JEE (Advanced) 2014, Paper-1, (3, 0)/60]
 (A) the first column of M is the transpose of the second row of M
 (B) the second row of M is the transpose of first column of M
 (C) M is a diagonal matrix with nonzero entries in the main diagonal
 (D) the product of entries in the main diagonal of M is not the square of an integer
- 17*. Let M and N be two 3×3 matrices such that $MN = NM$. Further, if $M \neq N^2$ and $M^2 = N^4$, then
[JEE (Advanced) 2014, Paper-1, (3, 0)/60]
 (A) determinant of $(M^2 + MN^2)$ is 0
 (B) there is a 3×3 non-zero matrix U such that $(M^2 + MN^2)U$ is the zero matrix
 (C) determinant of $(M^2 + MN^2) \geq 1$
 (D) for a 3×3 matrix U , if $(M^2 + MN^2)U$ equals the zero matrix then U is the zero matrix



- 18*. Let X and Y be two arbitrary, 3×3 , non-zero, skew-symmetric matrices and Z be an arbitrary 3×3 , non-zero, symmetric matrix. Then which of the following matrices is (are) skew symmetric ?

[JEE (Advanced) 2015, Paper-1 (4, -2)/88]

- (A) $Y^3Z^4 - Z^4Y^3$ (B) $X^{44} + Y^{44}$ (C) $X^4Z^3 - Z^3X^4$ (D) $X^{23} + Y^{23}$

- 19*. Which of the following values of α satisfy the equation
$$\begin{vmatrix} (1+\alpha)^2 & (1+2\alpha)^2 & (1+3\alpha)^2 \\ (2+\alpha)^2 & (2+2\alpha)^2 & (2+3\alpha)^2 \\ (3+\alpha)^2 & (3+2\alpha)^2 & (3+3\alpha)^2 \end{vmatrix} = -648\alpha$$
 ?

[JEE (Advanced) 2015, Paper-1 (4, -2)/88]

- (A) -4 (B) 9 (C) -9 (D) 4

20. Let $P = \begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & \alpha \\ 3 & -5 & 0 \end{bmatrix}$, where $\alpha \in \mathbb{R}$. Suppose $Q = [q_{ij}]$ is a matrix such that $PQ = kI$, where $k \in \mathbb{R}$, $k \neq 0$

and I is the identity matrix of order 3. If $q_{23} = -\frac{k}{8}$ and $\det(Q) = \frac{k^2}{2}$, then

[JEE (Advanced) 2016, Paper-1 (4, -2)/62]

- (A) $\alpha = 0, k = 8$ (B) $4\alpha - k + 8 = 0$ (C) $\det(P \operatorname{adj}(Q)) = 2^9$ (D) $\det(Q \operatorname{adj}(P)) = 2^{13}$

21. The total number of distinct $x \in \mathbb{R}$ for which
$$\begin{vmatrix} x & x^2 & 1+x^3 \\ 2x & 4x^2 & 1+8x^3 \\ 3x & 9x^2 & 1+27x^3 \end{vmatrix} = 10$$
 is

[JEE (Advanced) 2016, Paper-1 (3, 0)/62]

22. Let $P = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 16 & 4 & 1 \end{bmatrix}$ and I be the identity matrix of order 3. If $Q = [q_{ij}]$ is a matrix such that $P^{50} - Q = I$,

then $\frac{q_{31} + q_{32}}{q_{21}}$ equals

[JEE (Advanced) 2016, Paper-2 (3, -1)/62]

- (A) 52 (B) 103 (C) 201 (D) 205

23. Let $a, \lambda, \mu \in \mathbb{R}$. Consider the system of linear equations

$$ax + 2y = \lambda$$

$$3x - 2y = \mu$$

[JEE (Advanced) 2016, Paper-2 (4, -2)/62]

Which of the following statement(s) is(are) correct ?

- (A) if $a = -3$, then the system has infinitely many solutions for all values of λ and μ
 (B) If $a \neq -3$, then the system has a unique solution for all values of λ and μ
 (C) If $\lambda + \mu = 0$, then the system has infinitely many solutions for $a = -3$
 (D) If $\lambda + \mu \neq 0$, then the system has no solution for $a = -3$

24. Which of the following is(are) NOT the square of a 3×3 matrix with real entries?

[JEE(Advanced) 2017, Paper-1, (4, -2)/61]

- (A) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ (B) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ (C) $\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

25. For a real number α , if the system

[JEE(Advanced) 2017, Paper-1, (3, 0)/61]

$$\begin{bmatrix} 1 & \alpha & \alpha^2 \\ \alpha & 1 & \alpha \\ \alpha^2 & \alpha & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

of linear equations, has infinitely many solutions, then $1 + \alpha + \alpha^2 =$

26. How many 3×3 matrices M with entries from $\{0, 1, 2\}$ are there, for which the sum of the diagonal entries of $M^T M$ is 5 ?

[JEE(Advanced) 2017, Paper-2, (3, -1)/61]

- (A) 198 (B) 162 (C) 126 (D) 135





27. Let S be the set of all column matrices $\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$ such that $b_1, b_2, b_3 \in \mathbb{R}$ and the system of equation (in real variables)

$$\begin{aligned} -x + 2y + 5z &= b_1 \\ 2x - 4y + 3z &= b_2 \\ x - 2y + 2z &= b_3 \end{aligned}$$

has at least one solution. Then, which of the following system(s) (in real variables) has (have) at least

one solution for each $\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \in S$?

[JEE(Advanced) 2018, Paper-2, (4, -2)/60]

- (A) $x + 2y + 3z = b_1, 4y + 5z = b_2$ and $x + 2y + 6z = b_3$
 (B) $x + y + 3z = b_1, 5x + 2y + 6z = b_2$ and $-2x - y - 3z = b_3$
 (C) $-x + 2y - 5z = b_1, 2x - 4y + 10z = b_2$ and $x - 2y + 5z = b_3$
 (D) $x + 2y + 5z = b_1, 2x + 3z = b_2$ and $x + 4y - 5z = b_3$

28. Let P be a matrix of order 3×3 such that all the entries in P are from the set $\{-1, 0, 1\}$. Then, the maximum possible value of the determinant of P is _____. [JEE(Advanced) 2018, Paper-2, (3, 0)/60]

29. Let $M = \begin{bmatrix} \sin^4 \theta & -1 - \sin^2 \theta \\ 1 + \cos^2 \theta & \cos^4 \theta \end{bmatrix} = \alpha I + \beta M^{-1}$

Where $\alpha = \alpha(\theta)$ and $\beta = \beta(\theta)$ are real numbers and I is the 2×2 identity matrix.

If α^* is the minimum of the set $\{\alpha(\theta) : \theta \in [0, 2\pi]\}$ and

β^* is the minimum of the set $\{\beta(\theta) : \theta \in [0, 2\pi]\}$

[JEE(Advanced) 2019, Paper-1, (4, -1)/62]

Then the value of $\alpha^* + \beta^*$ is

- (A) $-\frac{37}{16}$ (B) $-\frac{29}{16}$ (C) $-\frac{31}{16}$ (D) $-\frac{17}{16}$

30. Let $M = \begin{bmatrix} 0 & 1 & a \\ 1 & 2 & 3 \\ 3 & b & 1 \end{bmatrix}$ and $\text{adj } M = \begin{bmatrix} -1 & 1 & -1 \\ 8 & -6 & 2 \\ -5 & 3 & -1 \end{bmatrix}$

where a and b are real numbers. Which of the following options is/are correct ?

(A) $\det(\text{adj } M^2) = 81$

(B) $a + b = 3$

(C) If $M \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, then $\alpha - \beta + \gamma = 3$

(D) $(\text{adj } M)^{-1} + \text{adj } M^{-1} = -M$

[JEE(Advanced) 2019, Paper-1, (4, -1)/62]

31. Let $x \in \mathbb{R}$ and let

[JEE(Advanced) 2019, Paper-2, (4, -1)/62]

$$P = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}, Q = \begin{bmatrix} 2 & x & x \\ 0 & 4 & 0 \\ x & x & 6 \end{bmatrix} \text{ and } R = P Q P^{-1}. \text{ Then which of the following is/are correct}$$

(A) there exists a real number x such that $PQ = QP$

(B) $\det R = \det \begin{bmatrix} 2 & x & x \\ 0 & 4 & 0 \\ x & x & 5 \end{bmatrix} + 8$ for all $x \in \mathbb{R}$

(C) For $x = 1$ there exists a unit vector $\alpha \hat{i} + \beta \hat{j} + \gamma \hat{k}$ for which are $R \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

(D) For $x = 0$ if $R \begin{bmatrix} 1 \\ a \\ b \end{bmatrix} = 6 \begin{bmatrix} 1 \\ a \\ b \end{bmatrix}$ then $a + b = 5$



32. Let $P_1 = I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $P_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, $P_3 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$,
 $P_4 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$, $P_5 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$, $P_6 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$,
and $X = \sum_{k=1}^6 P_k \begin{bmatrix} 2 & 1 & 3 \\ 1 & 0 & 2 \\ 3 & 2 & 1 \end{bmatrix} P_k^T$. [JEE(Advanced) 2019, Paper-2, (4, -1)/62]

Where P_k^T denotes the transpose of matrix P_k . Then which of the following options is/are correct ?

- (A) X is a symmetric matrix
 (B) The sum of diagonal entries of X is 18.

(C) If $X \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, then $\alpha = 30$

- (D) $X - 30I$ is an invertible matrix

PART - II : JEE (MAIN) / AIEEE PROBLEMS (PREVIOUS YEARS)

1. The number of 3×3 non-singular matrices, with four entries as 1 and all other entries as 0, is [AIEEE 2010 (8, -2), 144]
 (1) 5 (2) 6 (3) at least 7 (4) less than 4
2. Let A be a 2×2 matrix with non-zero entries and let $A^2 = I$, where I is 2×2 identity matrix. $\text{Tr}(A)$ = sum of diagonal elements of A and $|A|$ = determinant of matrix A. [AIEEE 2010 (4, -1), 144]
Statement -1 : $\text{Tr}(A) = 0$
Statement -2 : $|A| = 1$
 (1) Statement -1 is true, Statement-2 is true ; Statement -2 is not a correct explanation for Statement -1.
 (2) Statement-1 is true, Statement-2 is false.
 (3) Statement -1 is false, Statement -2 is true.
 (4) Statement -1 is true, Statement -2 is true; Statement-2 is a correct explanation for Statement-1.
3. Consider the system of linear equations : [AIEEE 2010 (4, -1), 144]
 $x_1 + 2x_2 + x_3 = 3$
 $2x_1 + 3x_2 + x_3 = 3$
 $3x_1 + 5x_2 + 2x_3 = 1$
 The system has
 (1) exactly 3 solutions (2) a unique solution (3) no solution (4) infinite number of solutions
4. Let A and B be two symmetric matrices of order 3. [AIEEE 2011, I, (4, -1), 120]
Statement-1 : $A(BA)$ and $(AB)A$ are symmetric matrices.
Statement-2 : AB is symmetric matrix if matrix multiplication of A with B is commutative.
 (1) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1.
 (2) Statement-1 is true, Statement-2 is true; Statement-2 is true; Statement-2 is **not** a correct explanation for Statement-1.
 (3) Statement-1 is true, Statement-2 is false.
 (4) Statement-1 is false, Statement-2 is true.



5. The number of values of k for which the linear equations
 $4x + ky + 2z = 0$
 $kx + 4y + z = 0$
 $2x + 2y + z = 0$
 posses a non-zero solution is :
 (1) 3 (2) 2 (3) 1 (4) zero [AIEEE 2011, I, (4, -1), 120]
6. If $\omega \neq 1$ is the complex cube root of unity and matrix $H = \begin{bmatrix} \omega & 0 \\ 0 & \omega \end{bmatrix}$, then H^{70} is equal to -
 (1) 0 (2) $-H$ (3) H^2 (4) H [AIEEE 2011, I, (4, -1), 120]
7. If the trivial solution is the only solution of the system of equations
 $x - ky + z = 0$
 $kx + 3y - kz = 0$
 $3x + y - z = 0$
 then the set of all values of k is :
 (1) $R - \{2, -3\}$ (2) $R - \{2\}$ (3) $R - \{-3\}$ (4) $\{2, -3\}$ [AIEEE 2011, II, (4, -1), 120]
8. **Statement - 1** : Determinant of a skew-symmetric matrix of order 3 is zero.
Statement - 2 : For any matrix A , $\det(A)^T = \det(A)$ and $\det(-A) = -\det(A)$.
 Where $\det(B)$ denotes the determinant of matrix B . Then :
 (1) Both statements are true (2) Both statements are false
 (3) Statement-1 is false and statement-2 is true. (4) Statement-1 is true and statement-2 is false [AIEEE 2011, II, (4, -1), 120]
9. Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{pmatrix}$. If u_1 and u_2 are column matrices such that $Au_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ and $Au_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, then $u_1 + u_2$ is equal to :
 (1) $\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$ (2) $\begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}$ (3) $\begin{pmatrix} -1 \\ -1 \\ 0 \end{pmatrix}$ (4) $\begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}$ [AIEEE-2012, (4, -1)/120]
10. Let P and Q be 3×3 matrices $P \neq Q$. If $P^3 = Q^3$ and $P^2Q = Q^2P$, then determinant of $(P^2 + Q^2)$ is equal to :
 (1) -2 (2) 1 (3) 0 (4) -1 [AIEEE-2012, (4, -1)/120]
11. The number of values of k , for which the system of equations :
 $(k + 1)x + 8y = 4k$
 $kx + (k + 3)y = 3k - 1$
 has no solution, is
 (1) infinite (2) 1 (3) 2 (4) 3 [AIEEE - 2013, (4, -1) 120]
12. If $P = \begin{bmatrix} 1 & \alpha & 3 \\ 1 & 3 & 3 \\ 2 & 4 & 4 \end{bmatrix}$ is the adjoint of a 3×3 matrix A and $|A| = 4$, then α is equal to :
 (1) 4 (2) 11 (3) 5 (4) 0 [AIEEE - 2013, (4, -1) 120]



13. If $\alpha, \beta \neq 0$ and $f(n) = \alpha^n + \beta^n$ and $\begin{vmatrix} 3 & 1+f(1) & 1+f(2) \\ 1+f(1) & 1+f(2) & 1+f(3) \\ 1+f(2) & 1+f(3) & 1+f(4) \end{vmatrix} = K (1-\alpha)^2 (1-\beta)^2 (\alpha-\beta)^2$, then K is equal to
- [JEE(Main) 2014, (4, -1), 120]
- (1) 1 (2) -1 (3) $\alpha\beta$ (4) $\frac{1}{\alpha\beta}$
14. If A is an 3×3 non-singular matrix such that $AA' = A'A$ and $B = A^{-1}A'$, then BB' equals :
- [JEE(Main) 2014, (4, -1), 120]
- (1) B^{-1} (2) $(B^{-1})'$ (3) $I + B$ (4) I
15. If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ a & 2 & b \end{bmatrix}$ is a matrix satisfying the equation $AA^T = 9I$, where I is 3×3 identity matrix, then the ordered pair (a, b) is equal to :
- [JEE(Main) 2015, (4, -1), 120]
- (1) (2, -1) (2) (-2, 1) (3) (2, 1) (4) (-2, -1)
16. The set of all value of λ for which the system of linear equations : [JEE(Main) 2015, (4, -1), 120]
- $$\begin{aligned} 2x_1 - 2x_2 + x_3 &= \lambda x_1 \\ 2x_1 - 3x_2 + 2x_3 &= \lambda x_2 \\ -x_1 + 2x_2 &= \lambda x_3 \end{aligned}$$
- has a non-trivial solution,
- (1) is an empty set (2) is a singleton
(3) contains two elements (4) contains more than two elements
17. The system of linear equations
- $$\begin{aligned} x + \lambda y - z &= 0 \\ \lambda x - y - z &= 0 \\ x + y - \lambda z &= 0 \end{aligned}$$
- has a non-trivial solution for :
- (1) Exactly one value of λ . (2) Exactly two values of λ .
(3) Exactly three values of λ . (4) Infinitely many values of λ .
18. If $A = \begin{bmatrix} 5a & -b \\ 3 & 2 \end{bmatrix}$ and $A \text{ adj } A = A A^T$, then $5a + b$ is equal to
- [JEE(Main) 2016, (4, -1), 120]
- (1) 5 (2) 4 (3) 13 (4) -1
19. It S is the set of distinct values of 'b' for which the following system of linear equations
- $$\begin{aligned} x + y + z &= 1 \\ x + ay + z &= 1 \\ ax + by + z &= 0 \end{aligned}$$
- [JEE(Main) 2017, (4, -1), 120]
- has no solution, then S is :
- (1) an empty set (2) an infinite set
(3) a finite set containing two or more elements (4) a singleton
20. If $A = \begin{bmatrix} 2 & -3 \\ -4 & 1 \end{bmatrix}$, then $\text{adj } (3A^2 + 12A)$ is equal to
- [JEE(Main) 2017, (4, -1), 120]
- (1) $\begin{bmatrix} 72 & -84 \\ -63 & 51 \end{bmatrix}$ (2) $\begin{bmatrix} 51 & 63 \\ 84 & 72 \end{bmatrix}$ (3) $\begin{bmatrix} 51 & 84 \\ 63 & 72 \end{bmatrix}$ (4) $\begin{bmatrix} 72 & -63 \\ -84 & 51 \end{bmatrix}$



21. If the system of linear equations

$$x + ky + 3z = 0$$

$$3x + ky - 2z = 0$$

$$2x + 4y - 3z = 0$$

has a non-zero solution (x, y, z) , then $\frac{xz}{y^2}$ is equal to :

(1) -30

(2) 30

(3) -10

(4) 10

[JEE(Main) 2018, (4, -1), 120]

22. If $\begin{bmatrix} x-4 & 2x & 2x \\ 2x & x-4 & 2x \\ 2x & 2x & x-4 \end{bmatrix} = (A + Bx)(x-A)^2$ then the ordered pair (A, B) is equal to :

(1) $(-4, 5)$

(2) $(4, 5)$

(3) $(-4, -5)$

(4) $(-4, 3)$

[JEE(Main) 2018, (4, -1), 120]

23. The system of linear equations

$$x + y + z = 2$$

$$2x + 3y + 2z = 5$$

$$2x + 3y + (a^2 - 1)z = a + 1$$

(1) is inconsistent when $a = 4$

(2) has infinitely many solutions for $a = 4$

(3) is inconsistent when $|a| = \sqrt{3}$

(4) has a unique solution for $|a| = \sqrt{3}$

[JEE(Main) 2019, Online (09-01-19), P-1 (4, -1), 120]

24. If $A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$, then the matrix A^{-50} when $\theta = \frac{\pi}{12}$, is equal to :

(1) $\begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$

(2) $\begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$

(3) $\begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$

(4) $\begin{bmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$

[JEE(Main) 2019, Online (09-01-19), P-1 (4, -1), 120]

25. Let $d \in \mathbb{R}$, and $A = \begin{bmatrix} -2 & 4+d & (\sin \theta - 2) \\ 1 & (\sin \theta) + 2 & d \\ 5 & (2 \sin \theta) - d & (-\sin \theta) + 2 + 2d \end{bmatrix}$, $\theta \in [0, 2\pi]$. If the minimum value of $\det(A)$ is 8, then a value of d is :

(1) -5

(2) $2(\sqrt{2} + 2)$

(3) $2(\sqrt{2} + 1)$

(4) -7

[JEE(Main) 2019, Online (10-01-19), P-1 (4, -1), 120]

26. Let $A = \begin{pmatrix} 0 & 2q & r \\ p & q & -r \\ p & -q & r \end{pmatrix}$. If $AA^T = I_3$, then $|p|$ is

(1) $\frac{1}{\sqrt{5}}$

(2) $\frac{1}{\sqrt{3}}$

(3) $\frac{1}{\sqrt{6}}$

(4) $\frac{1}{\sqrt{2}}$

[JEE(Main) 2019, Online (11-01-19), P-1 (4, -1), 120]

27. Let $A = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}$, $(\alpha \in \mathbb{R})$ such that $A^{32} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Then a value of α is :

(1) $\frac{\pi}{16}$

(2) $\frac{\pi}{32}$

(3) 0

(4) $\frac{\pi}{64}$

[JEE(Main) 2019, Online (08-04-19), P-1 (4, -1), 120]

28. If $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \cdots \begin{bmatrix} 1 & n-1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 78 \\ 0 & 1 \end{bmatrix}$, then the inverse of $\begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$ is :

(1) $\begin{bmatrix} 1 & 0 \\ 12 & 1 \end{bmatrix}$

(2) $\begin{bmatrix} 1 & 0 \\ 13 & 1 \end{bmatrix}$

(3) $\begin{bmatrix} 1 & -12 \\ 0 & 1 \end{bmatrix}$

(4) $\begin{bmatrix} 1 & -13 \\ 0 & 1 \end{bmatrix}$

[JEE(Main) 2019, Online (09-04-19), P-1 (4, -1), 120]





29. If A is a symmetric matrix and B is skew symmetric matrix such that

$$A + B = \begin{bmatrix} 2 & 3 \\ 5 & -1 \end{bmatrix}, \text{ then } AB \text{ is equal to}$$

[JEE(Main) 2019, Online (12-04-19), P-1 (4, -1), 120]

(1) $\begin{bmatrix} -4 & 2 \\ 1 & 4 \end{bmatrix}$

(2) $\begin{bmatrix} 4 & -2 \\ -1 & -4 \end{bmatrix}$

(3) $\begin{bmatrix} 4 & -2 \\ 1 & -4 \end{bmatrix}$

(4) $\begin{bmatrix} -4 & -2 \\ -1 & 4 \end{bmatrix}$

30. The total number of matrices $A = \begin{pmatrix} 0 & 2y & 1 \\ 2x & y & -1 \\ 2x & -y & 1 \end{pmatrix}$, $(x, y \in \mathbb{R}, x \neq y)$ for which $A^T A = 3I_3$ is :

[JEE(Main) 2019, Online (09-04-19), P-2 (4, -1), 120]

(1) 3

(2) 4

(3) 6

(4) 2

31. If the system of equations $2x + 3y - z = 0$, $x + ky - 2z = 0$ and $2x - y + z = 0$ has a non-trivial solution (x, y, z) , then $\frac{x}{y} + \frac{y}{z} + \frac{z}{x} + k$ is equal to :

[JEE(Main) 2019, Online (09-04-19), P-2 (4, -1), 120]

(1) -4

(2) $\frac{3}{4}$

(3) $\frac{1}{2}$

(4) $-\frac{1}{4}$

32. Let α be a root of equation $x^2 + x + 1 = 0$ and the matrix $A = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \alpha^2 & \alpha^4 \end{bmatrix}$, then the matrix A^{31} is

equal to :

[JEE(Main) 2020, Online (07-01-20), P-1 (4, -1), 120]

(1) A^3

(2) A^2

(3) I_3

(4) A

33. Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be two 3×3 real matrices such that $b_{ij} = (3)^{(i+j-2)} a_{ij}$, where $i, j = 1, 2, 3$. If the determinant of B is 81, then the determinant of A is:

[JEE(Main) 2020, Online (07-01-20), P-2 (4, -1), 120]

(1) $1/9$

(2) $1/81$

(3) 3

(4) $1/3$



Answers

EXERCISE - 1

PART - I

Section (A):

A-1. $\begin{bmatrix} 1 & 0 \\ 3 & 2 \\ 5 & 4 \end{bmatrix}$

A-2. $(x, y, z, w) = (1, 2, 4, 5)$

A-3. $\begin{bmatrix} 2 & -1/2 \\ 4 & -1 \end{bmatrix}$

A-4. $AB = \begin{bmatrix} 18 & -11 & 10 \\ -16 & 47 & 10 \\ 62 & -23 & 42 \end{bmatrix}, BA = \begin{bmatrix} 49 & 24 \\ -7 & 58 \end{bmatrix}$

A-7. $y \in R$

A-9. Zero

Section (B) :

B-1. $0, \frac{3\pi}{4}, \pi$

B-2. (i) 0 (ii) 0 (iii) 0 (iv) $5(3\sqrt{2} - 5\sqrt{3})$

B-5. (i) $x = -2b/a$ (ii) 4

B-8. $A = 0, B = 0$

Section (C) :

C-1. $\begin{bmatrix} -191 & -110 \\ 77 & 44 \end{bmatrix}$

C-2. (ii) $|A|^{(n-1)^3}$

C-3. $\begin{bmatrix} 9 & -3 & 5 \\ -2 & 1 & 0 \\ 1 & 0 & 2 \end{bmatrix}$

C-5. $a = 1, c = -1$

Section (D) :

D-1. $a = -4, b = 1, A^{-1} = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}$

D-2. 0

D-4. (i) $x = 3, y = 4, z = 6$ (ii) $x = -\frac{5k}{3} + \frac{8}{3}, y = -\frac{2k}{3} - \frac{1}{3}, z = k$, where $k \in R$

D-5. $x = -7, y = -4$

D-6. for $c = 0, x = -3, y = 3$; for $c = -10, x = -\frac{1}{2}, y = \frac{4}{3}$

D-7. (i) $x = 2, y = 2, z = 2$ (ii) $x = 1, y = 3, z = 5$

D-8. (a) $\lambda \neq 3$ (b) $\lambda = 3, \mu = 10$ (c) $\lambda = 3, \mu \neq 10$

D-9. $x = 3, y = -2, z = -1$

D-10. $x = 1, y = 2, z = 3, A^{-1} = \frac{1}{17} \begin{bmatrix} 1 & 5 & 1 \\ 8 & 6 & -9 \\ 10 & -1 & -7 \end{bmatrix}$

D-11. S1, S3, S4

PART - II

Section (A):

A-1. (A) A-2. (D) A-3. (A) A-4. (A) A-5. (C) A-6. (B) A-7. (C)
A-8. (A) A-9. (A) A-10. (A) A-11. (A)

Section (B) :

B-1. (B) B-2. (A) B-3. (D) B-4. (D) B-5. (D) B-6. (B) B-7. (C)
B-8. (B) B-9. (B)

**Section (C) :**

C-1. (A) C-2. (C) C-3. (A) C-4. (A) C-5. (C) C-6. (C)

Section (D) :

D-1. (A) D-2. (A) D-3. (D) D-4. (B) D-5. (C)

PART - III

1. (A) → (s), (B) → (p), (C) → (p), (D) → (p)
 2. (A) → (q), (B) → (p), (C) → (s), (D) → (q)

EXERCISE - 2**PART - I**

1. (D) 2. (D) 3. (C) 4. (D) 5. (A) 6. (C) 7. (A)
 8. (A) 9. (A) 10. (A) 11. (B) 12. (A) 13. (A) 14. (A)
 15. (C) 16. (B) 17. (B) 18. (A) 19. (A) 20. (A) 21. (B)
 22. (A)

PART - II

1. 02.00 2. 37.03 or 37.04 3. 36.00 4. 00.50 5. 02.38 6. 16.00
 7. 00.00 8. 63.00 9. 04.00 10. 22.92 or 22.93 11. 02.00
 12. 01.00 13. 03.14 14. 80.00 15. 39.00 16. 12.20 17. 08.00

PART - III

1. (ABCD) 2. (ABD) 3. (AB) 4. (ABC) 5. (BD) 6. (AC)
 7. (BCD) 8. (BCD) 9. (ABD) 10. (BD) 11. (AD) 12. (BC)
 13. (ABD) 14. (AC) 15. (BCD) 16. (AB) 17. (ABC) 18. (BC)

PART - IV

1. (AC) 2. (AC) 3. (AC) 4. (A) 5. (D) 6. (C)

EXERCISE - 3**PART - I**

1. (D) 2. (C) 3. (D) 4. 3 5. 4
 6. C (In JEE this question was bonus because in JEE instead of $2n \times 2n$, 3×3 was given and we know that there is no non-singular 3×3 skew symmetric matrix).
 7. (D) 8. (A) 9. (B) 10. (A) 11. 9 12. (D) 13. (D)
 14. (AD) 15. (CD) 16. (CD) 17. (AB) 18. (C,D) 19. (B,C) 20. (B,C)
 21. 2 22. (B) 23. (B,C,D) 24. (A,C) 25. (1) 26. (A) 27. (AD)
 28. (4) 29. (B) 30. (BCD) 31. (BD) 32. (ABC)

PART - II

1. (3) 2. (2) 3. (3) 4. (2) 5. (2) 6. (4) 7. (1)
 8. (4) 9. (4) 10. (3) 11. (2) 12. (2) 13. (1) 14. (4)
 15. (4) 16. (3) 17. (3) 18. (1) 19. (4) 20. (2) 21. (4)
 22. (1) 23. (3) 24. (2) 25. (1) 26. (4) 27. (4) 28. (4)
 29. (2) 30. (2) 31. (3) 32. (1) 33. (1)



High Level Problems (HLP)

- If $a^2 + b^2 + c^2 = 1$, then prove that

$$\begin{vmatrix} a^2 + (b^2 + c^2)\cos\phi & ab(1 - \cos\phi) & ac(1 - \cos\phi) \\ ba(1 - \cos\phi) & b^2 + (c^2 + a^2)\cos\phi & bc(1 - \cos\phi) \\ ca(1 - \cos\phi) & cb(1 - \cos\phi) & c^2 + (a^2 + b^2)\cos\phi \end{vmatrix}$$

is independent of a, b, c
- If $a, b, c, x, y, z \in \mathbb{R}$, then prove that,

$$\begin{vmatrix} (a-x)^2 & (b-x)^2 & (c-x)^2 \\ (a-y)^2 & (b-y)^2 & (c-y)^2 \\ (a-z)^2 & (b-z)^2 & (c-z)^2 \end{vmatrix} = \begin{vmatrix} (1+ax)^2 & (1+bx)^2 & (1+cx)^2 \\ (1+ay)^2 & (1+by)^2 & (1+cy)^2 \\ (1+az)^2 & (1+bz)^2 & (1+cz)^2 \end{vmatrix}$$
- If a_1, a_2, a_3 are distinct real roots of the equation $px^3 + px^2 + qx + r = 0$ such that

$$\begin{vmatrix} 1+a_1 & 1 & 1 \\ 1 & 1+a_2 & 1 \\ 1 & 1 & 1+a_3 \end{vmatrix} = 0$$

then Prove that $\frac{r}{p} < 0$
- Prove that $\Delta = \begin{vmatrix} \beta\gamma & \beta\gamma' + \beta'\gamma & \beta'\gamma' \\ \gamma\alpha & \gamma\alpha' + \gamma'\alpha & \gamma'\alpha' \\ \alpha\beta & \alpha\beta' + \alpha'\beta & \alpha'\beta' \end{vmatrix} = (\alpha\beta' - \alpha'\beta)(\beta\gamma' - \beta'\gamma)(\gamma\alpha' - \gamma'\alpha)$
- If

$$\begin{aligned} ax_1^2 + by_1^2 + cz_1^2 &= d \\ ax_2^2 + by_2^2 + cz_2^2 &= d \\ ax_3^2 + by_3^2 + cz_3^2 &= d \end{aligned} \quad \text{and} \quad \begin{aligned} ax_2x_3 + by_2y_3 + cz_2z_3 &= f \\ ax_3x_1 + by_3y_1 + cz_3z_1 &= f \\ ax_1x_2 + by_1y_2 + cz_1z_2 &= f \end{aligned}$$

then prove that $\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix}^2 = (d-f)^2 \frac{d+2f}{abc}$, where $a, b, c \neq 0$.
- If $(x_1 - x_2)^2 + (y_1 - y_2)^2 = a^2$
 $(x_2 - x_3)^2 + (y_2 - y_3)^2 = b^2$, prove that $4 \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}^2 = (a+b+c)(b+c-a)(c+a-b)(a+b-c)$
 $(x_3 - x_1)^2 + (y_3 - y_1)^2 = c^2$
- Let $A = \begin{bmatrix} \cos^{-1}x & \cos^{-1}y & \cos^{-1}z \\ \cos^{-1}y & \cos^{-1}z & \cos^{-1}x \\ \cos^{-1}z & \cos^{-1}x & \cos^{-1}y \end{bmatrix}$ such that $|A| = 0$, then find the maximum value of $x + y + z$
- If $y = \frac{u}{v}$, where u & v are functions of ' x ', show that, $v^3 \frac{d^2y}{dx^2} = \begin{vmatrix} u & v & 0 \\ u' & v' & v \\ u'' & v'' & 2v' \end{vmatrix}$.
- If α, β be the real roots of $ax^2 + bx + c = 0$ and $s_n = \alpha^n + \beta^n$, then prove that $as_n + bs_{n-1} + cs_{n-2} = 0$ for all $n \geq 2, n \in \mathbb{N}$. Hence or otherwise prove that $\begin{vmatrix} 3 & 1+s_1 & 1+s_2 \\ 1+s_1 & 1+s_2 & 1+s_3 \\ 1+s_2 & 1+s_3 & 1+s_4 \end{vmatrix} \geq 0$ for all real a, b, c .





10. Let $a > 0, d > 0$. Find the value of determinant
$$\begin{vmatrix} \frac{1}{a} & \frac{1}{a(a+d)} & \frac{1}{(a+d)(a+2d)} \\ \frac{1}{(a+d)} & \frac{1}{(a+d)(a+2d)} & \frac{1}{(a+2d)(a+3d)} \\ \frac{1}{(a+2d)} & \frac{1}{(a+2d)(a+3d)} & \frac{1}{(a+3d)(a+4d)} \end{vmatrix}$$
11. Let $\vec{a}_r = x_r \hat{i} + y_r \hat{j} + z_r \hat{k}$, $r = 1, 2, 3$ be three mutually perpendicular unit vectors, then find the value of
$$\begin{vmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{vmatrix}$$
12. If
$$\begin{vmatrix} x^k & x^{k+2} & x^{k+3} \\ y^k & y^{k+2} & y^{k+3} \\ z^k & z^{k+2} & z^{k+3} \end{vmatrix} = (x-y)(y-z)(z-x) \left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z} \right)$$
, then find the value of k .
13. If the determinant
$$\begin{vmatrix} a+p & \ell+x & u+f \\ b+q & m+y & v+g \\ c+r & n+z & w+h \end{vmatrix}$$
 splits into exactly K determinants of order 3, each element of which contains only one term, then find the value of $K =$
14. If a, b, c are all different and
$$\begin{vmatrix} a & a^3 & a^4-1 \\ b & b^3 & b^4-1 \\ c & c^3 & c^4-1 \end{vmatrix} = 0$$
, then find the value of $abc(ab+bc+ca) - (a+b+c)$.
15. If a, b, c are complex numbers and $z = \begin{vmatrix} 0 & -b & -c \\ \bar{b} & 0 & -a \\ \bar{c} & \bar{a} & 0 \end{vmatrix}$ then show that z is purely imaginary.
16. If $f(x) = \log_{10} x$ and $g(x) = e^{\pi x}$ and $h(x) = \begin{vmatrix} f(x)g(x) & [f(x)]^{g(x)} & 1 \\ f(x^2)g(x^2) & [f(x^2)]^{g(x^2)} & 0 \\ f(x^3)g(x^3) & [f(x^3)]^{g(x^3)} & 1 \end{vmatrix}$, then find the value of $h(10)$.
17. If a, b, c , are real numbers, and $D = \begin{vmatrix} a & 1+2i & 3-5i \\ 1-2i & b & -7-3i \\ 3+5i & -7+3i & c \end{vmatrix}$ then show that D is purely real.
18. If $A = \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix}$ then find $\lim_{n \rightarrow \infty} \frac{1}{n} A^n$
19. Let $P = \begin{bmatrix} \cos \frac{\pi}{9} & \sin \frac{\pi}{9} \\ -\sin \frac{\pi}{9} & \cos \frac{\pi}{9} \end{bmatrix}$ and α, β, γ be non-zero real numbers such that $\alpha p^6 + \beta p^3 + \gamma I$ is the zero matrix. Then find value of $(\alpha^2 + \beta^2 + \gamma^2)^{(\alpha-\beta)(\beta-\gamma)(\gamma-\alpha)}$
20. Consider an odd order square symmetric matrix $A = [a_{ij}]_{n \times n}$. Its element in any row are 1, 2,, n in some order, then prove that $a_{11}, a_{22}, \dots, a_{nn}$ are numbers 1, 2, 3,, n in some order.



21. Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$; $B = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$ and $C = 3A + 7B$

Prove that

(i) $(A + B)^{2013} = A^{2013} + B^{2013}$

(ii) Prove that $A^n = 3^{n-1} A$; $B^n = 3^{n-1} B$; $C^n = 3^{2n-1} A + 7 \cdot 21^{n-1} B$.

22. Let 'A' is (4×4) matrix such that sum of elements in each row is 1. Find out sum of all the elements in A^{10} .

23. Let $A = \begin{bmatrix} x + \lambda & x & x \\ x & x + \lambda & x \\ x & x & x + \lambda \end{bmatrix}$, then prove that A^{-1} exists if $3x + \lambda \neq 0$, $\lambda \neq 0$

24. Prove that if A and B are $n \times n$ matrices, then $\det(I_n - AB) = \det(I_n - BA)$.

25. Let A be an $n \times n$ matrix such that $A^n = \alpha A$ where α is a real number different from 1 and -1 . Prove that the matrix $A + I_n$ is invertible.

26. Let p and q be real numbers such that $x^2 + px + q \neq 0$ for every real number x. Prove that if n is an odd positive integer, then $X^2 + pX + qI_n \neq 0_n$ for all real matrices X of order $n \times n$.

27. Let A, B, C be three 3×3 matrices with real entries. If $BA + BC + AC = I$ and $\det(A + B) = 0$ then find the value of $\det(A + B + C - BAC)$.

28. If $|z_1| = |z_2| = 1$, then prove that $\begin{bmatrix} z_1 & -z_2 \\ \bar{z}_2 & \bar{z}_1 \end{bmatrix}^{-1} \begin{bmatrix} \bar{z}_1 & z_2 \\ -\bar{z}_2 & z_1 \end{bmatrix}^{-1} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$

29. If A and B are two square matrices such that $B = -A^{-1}BA$, then show that $(A + B)^2 = A^2 + B^2$

30. If the system of equations $x = cy + bz$, $y = az + cx$ and $z = bx + ay$ has a non-zero solution and at least one of a, b, c is a proper fraction, prove that $a^3 + b^3 + c^3 < 3$ and $abc > -1$.

31. If $D = \text{diag}\{d_1, d_2, \dots, d_n\}$, then prove that $f(D) = \text{diag}\{f(d_1), f(d_2), \dots, f(d_n)\}$, where $f(x)$ is a polynomial with scalar coefficient.

32. Given the matrix $A = \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 5 \end{bmatrix}$ and X be the solution set of the equation $A^x = A$, where

$x \in \mathbf{N} - \{1\}$. Evaluate $\prod \frac{x^3 + 1}{x^3 - 1}$; where the continued product extends $\forall x \in X$.

Comprehension (Q. NO. 33 to 35)

Any non-zero vector, X, is said to be characteristic vector of a matrix A, if there exist a number λ such that $AX = \lambda X$. And then λ is said to be a characteristic root of the matrix A corresponding to the characteristic vector X and vice versa.

Also $AX = \lambda X \Rightarrow (A - \lambda I)X = 0$

Since $X \neq 0 \Rightarrow |A - \lambda I| = 0$

Thus every characteristic root λ of a matrix A is a root of its characteristic equation.

33. Prove that the two matrices A and $P^{-1}AP$ have the same characteristic roots and hence show that square matrices AB & BA have same characteristic roots if at least one of them is invertible.

34. If q is a characteristic root of a non singular matrix A, then prove that $\frac{|A|}{q}$ is a characteristic root of $\text{adj } A$.





35. Show that if $\alpha_1, \alpha_2, \dots, \alpha_n$ are n characteristic roots of a square matrix A of order n , then the roots of the matrix A^2 be $\alpha_1^2, \alpha_2^2, \dots, \alpha_n^2$.
36. If there are three square matrices A, B, C of same order satisfying the equation $A^3 = A^{-1}$ and let $B = -A^{3^n}$ and $C = A^{3^{(n+4)}}$ then prove that $\det(B + C) = 0, n \in \mathbb{N}$
37. If A is a non-singular matrix satisfying $AB - BA = A$, then prove that $\det.(B + I) = \det.(B - I)$
38. If rank is a number associated with a matrix which is the highest order of non-singular sub matrix then prove that
- (i) Rank of the matrix $A = \begin{bmatrix} 1 & 3 & -2 \\ 4 & -1 & 0 \\ 2 & -7 & 4 \end{bmatrix}$ is 2
- (ii) If the matrix $A = \begin{bmatrix} y+a & b & c \\ a & y+b & c \\ a & b & y+c \end{bmatrix}$ has rank 3, then $y \neq -(a + b + c)$ and $y \neq 0$
- (iii) If A & B are two square matrices of order 3 such that rank of matrix AB is two, then atleast one of A & B is singular.

Answers

7. 3 10. $\frac{4d^4}{a(a+d)^2(a+2d)^3(a+3d)^2(a+4d)}$ 11. ± 1 12. $k = -1$
13. 8 14. 0 16. 0 18. $\begin{bmatrix} 0 & a \\ 0 & 0 \end{bmatrix}$ 19. 1
22. 4 27. 0

